IOT SOLUTION DESIGN

Design of Enterprise IoT: Assignment 1b

OBJECTIVE

This assignment focuses on the **design phase of an IoT solution**.

Based on the previously defined use case, solution overview and value proposition in **Assignment 1a**, you are required to make informed design decisions across several key technical components. Specifically, you will select appropriate IoT devices (Task 2), determine suitable network technologies (Task 3), design how the system connects to and utilises edge/fog and cloud computing environments (Task 4), and develop a graphical connectivity diagram of the proposed solution (Task 5).

Note that **using a fully integrated existing solution will result in a zero mark** for this assignment. **Examples** include solutions that rely on a single product with all-in-one, built-in smart functionality, such as a mobile smart device with fully integrated features or a vehicle with entirely embedded technologies.

TASK 1: SOLUTION OVERVIEW (MAX 150 WORDS) [2 MARKS]

Provide an overview of the IoT solution plan proposed in **Assignment 1a**, incorporating refinements if necessary. It should highlight the key features of the solution to inform design decisions in the subsequent tasks.

TASK 2: IOT DEVICES [10 MARKS]

Using the requirements informed by your solution plan (especially features) in Task 1, identify commercially available devices (including sensors, actuators, and other IoT devices) suitable for your solution.

2.1. Provide an overview of your selection and briefly explain why these devices are suitable for your solution. **(MAX 100 WORDS)**

2.2. Present your selected devices using the provided template (**Template-2_IoT-Device-Profile.docx**). Ensure all selected devices are properly referenced with relevant URLs or in-text citations. **Failure to use the template will result in a zero mark for this task**.

Note that: (a) focus on off-the-shelf devices that are commercially available and ready to use; and (b) the use of smart mobile devices (e.g., smartphones) as the proposed solution is not permitted, except as a user interface.

TASK 3: IOT NETWORK TECHNOLOGIES [8 MARKS]

Based on the list of IoT devices specified in Task 2, identify the network technologies that are suitable to support IoT connectivity for your solution. Present your selected network technologies using the provided template (**Template-3_IoT-Connectivity.docx**). **Failure to use the template will result in a zero mark for this task**.

TASK 4: FOG AND CLOUD COMPUTING [10 MARKS]

Continue from the network technologies specified in Task 3 to determine the connectivity between IoT devices, network devices within the edge/fog layer and the cloud. Complete your connectivity proposal by including the edge/fog layer and cloud using the provided template (**Template-3_IoT-Connectivity.docx**). **Failure to use the template will result in a zero mark for this task**.

TASK 5: IOT CONNECTIVITY DIAGRAM [6 MARKS]

5.1. Present your overall IoT solution in an IoT connectivity diagram.

- Create a graphical representation of your overall IoT solution.
- The diagram should clearly illustrate the connectivity between IoT devices, network devices, the edge/fog layer, and the cloud, showing the flow of information and interaction within user applications and/or existing systems.
- Ensure that the diagram is clear, well-labelled, and includes relevant communication and network technologies. It should be easy to interpret and follow.
- You can use any drawing software or tool of your preference to create the diagram.

5.2. Describe the above IoT connectivity diagram to illustrate your overall IoT solution design. **(MAX 150 WORDS)**

TASK 6: CONCLUSION (MAX 150 WORDS) [2 MARKS]

Include a **Conclusion** section that:

- Summarises how effectively your solution design reflects the IoT solution plan in Task 1.
- Discusses any limitations of the proposed solution, highlighting what it cannot do or situations it cannot address.

REPORT PRESENTATION AND FORMATTING (2 MARKS)

- Your report should follow a well-structured format, presenting tasks in the specified order. Each section should clearly link to the previous (and possibly subsequent) sections, referencing relevant information from other sections where appropriate.
- Ensure the report is easy to follow, professionally presented, and free of language errors. It should include elements such as a cover page, table of contents, in-text citations, and a reference list, with fluent and clear language throughout.
- Present tables and diagrams clearly, providing brief explanations and, if applicable, proper APA 7 citations.
- Include sufficient references for relevant tasks, demonstrating effective use of cited sources.
- Use APA 7 style for in-text citations and the reference list.